

The 3^k Factorial Design

- k factors, each at 3 levels, totally 3^k treatment combinations.
- Factor levels can be denoted as 0, 1, 2 or 1, 2, 3, or -1, 0, 1.
- For a categorical factor, to compare three levels of the factor
- For a numerical factor, to study curvature in the response surface

Examples of Designs

3^2 Design

<i>A</i>	<i>B</i>
0	0
0	1
0	2
1	0
1	1
1	2
2	0
2	1
2	2

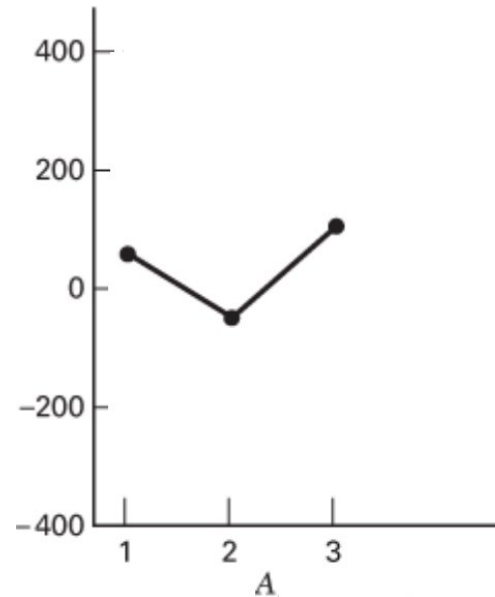
3^3 Design

Run	Factor		
	<i>A</i>	<i>B</i>	<i>C</i>
1	0	0	0
2	0	0	1
3	0	0	2
4	0	1	0
5	0	1	1
6	0	1	2
7	0	2	0
8	0	2	1
9	0	2	2
10	1	0	0
11	1	0	1
12	1	0	2
13	1	1	0
14	1	1	1
15	1	1	2
16	1	2	0
17	1	2	1
18	1	2	2
19	2	0	0
20	2	0	1
21	2	0	2
22	2	1	0
23	2	1	1
24	2	1	2
25	2	2	0
26	2	2	1
27	2	2	2

Main Effects

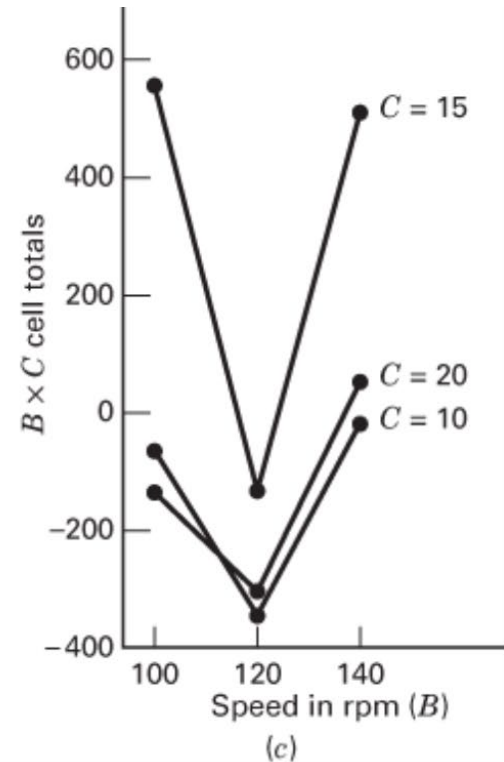
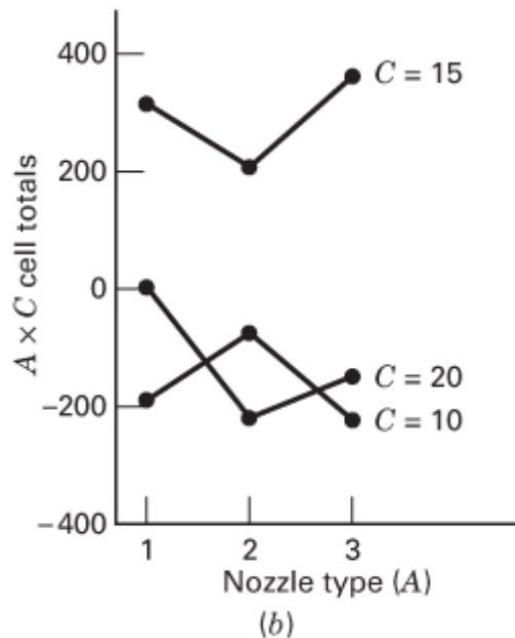
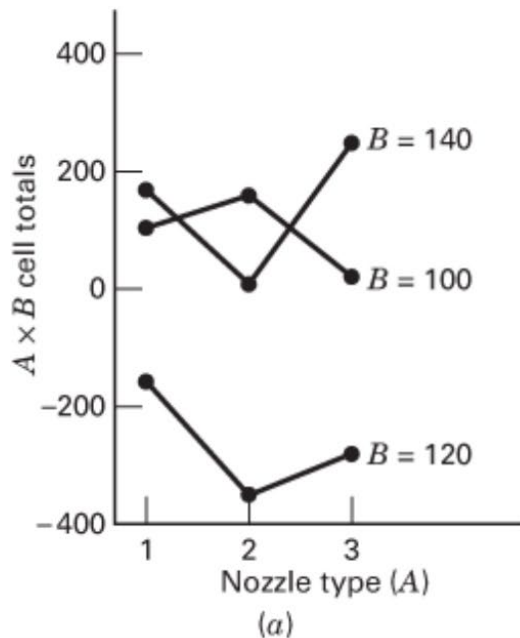
- Two degrees of freedom

- $\bar{y}_{A1} - \bar{y}_{A0}$
- $\bar{y}_{A2} - \bar{y}_{A1}$



Interaction Effects

- Two factor interaction: 4 degrees of freedom



EXAMPLE 9.1

A machine is used to fill 5-gallon metal containers with soft drink syrup. The variable of interest is the amount of syrup loss due to frothing. Three factors are thought to influence frothing: the nozzle design (*A*), the filling speed (*B*), and the operating pressure (*C*). Three nozzles, three filling speeds, and three pressures are chosen, and two replicates of a 3^3 factorial experiment are run. The coded data are shown in Table 9.1.

The analysis of variance for the syrup loss data is shown in Table 9.2. The sums of squares have been

computed by the usual methods. We see that the filling speed and operating pressure are statistically significant. All three two-factor interactions are also significant. The two-factor interactions are analyzed graphically in Figure 9.4. The middle level of speed gives the best performance, nozzle types 2 and 3, and either the low (10 psi) or high (20 psi) pressure seems most effective in reducing syrup loss.

■ TABLE 9.1

Syrup Loss Data for Example 9.1 (units are cubic centimeters –70)

Pressure (in psi) (C)	Nozzle Type (A)								
	1			2			3		
	Speed (in RPM) (B)								
	100	120	140	100	120	140	100	120	140
10	−35	−45	−40	17	−65	20	−39	−55	15
	−25	−60	15	24	−58	4	−35	−67	−30
15	110	−10	80	55	−55	110	90	−28	110
	75	30	54	120	−44	44	113	−26	135
20	4	−40	31	−23	−64	−20	−30	−61	54
	5	−30	36	−5	−62	−31	−55	−52	4

ANOVA Model

An example with 3 factors

$$Y_{ijkl} = \mu + \alpha_i + \beta_j + \gamma_k + (\alpha\beta)_{ij} + (\alpha\gamma)_{ik} + (\beta\gamma)_{jk} + (\alpha\beta\gamma)_{ijk} + \varepsilon_{ijkl}$$

- $i, j, k \in \{1, 2, 3\}, \ell = 1, \dots, r$
 - $\varepsilon_{ijkl} \sim N(0, \sigma^2)$
-

Constraints (sum-to-zero for identifiability):

$$\sum_i \alpha_i = 0, \quad \sum_j \beta_j = 0, \quad \sum_k \gamma_k = 0$$

$$\sum_i (\alpha\beta)_{ij} = 0, \quad \sum_j (\alpha\beta)_{ij} = 0$$

(similarly for other interactions)

ANOVA Decomposition

Total variation:

$$SS_T = SS_A + SS_B + SS_C + SS_{AB} + SS_{AC} + SS_{BC} + SS_{ABC} + SS_E$$

Degrees of freedom:

- Each factor: 2 df
 - Each 2-way interaction: 4 df
 - 3-way interaction: 8 df
 - Error: $N - 27$
-
- Hypothesis

Main effects

$$H_0^A : \alpha_1 = \alpha_2 = \alpha_3 = 0$$

Interaction effects

$$H_0^{AB} : (\alpha\beta)_{ij} = 0 \forall i, j$$

ANOVA

TABLE 9.2

Analysis of Variance for Syrup Loss Data

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P -Value
<i>A</i> , nozzle	993.77	2	496.89	1.17	0.3256
<i>B</i> , speed	61,190.33	2	30,595.17	71.74	< 0.0001
<i>C</i> , pressure	69,105.33	2	34,552.67	81.01	< 0.0001
<i>AB</i>	6,300.90	4	1,575.22	3.69	0.0160
<i>AC</i>	7,513.90	4	1,878.47	4.40	0.0072
<i>BC</i>	12,854.34	4	3,213.58	7.53	0.0003
<i>ABC</i>	4,628.76	8	578.60	1.36	0.2580
Error	11,515.50	27	426.50		
Total	174,102.83	53			

df breakdown:

$$abn - 1 = a - 1 + b - 1 + (a - 1)(b - 1) + ab(n - 1)$$

- # Enter the 54 responses
- `y <- c(`
- `-35,-25,-45,-60,-40,15, 110,75,-10,30,80,54, 4,5,-40,-30,31,36,`
- `17,24,-65,-58,20,4, 55,120,-55,-44,110,44, -31,-5,-64,-62,-20,-31,`
- `-39,-35,-55,-67,15,-30, 90,113,-28,-26,110,135, -30,-55,-61,-52,54,4`
- `)`
- `A <- factor(rep(1:3, each = 18))`
- `C <- factor(rep(rep(c(10,15,20), each = 6), times = 3))`
- `B <- factor(rep(rep(c(100,120,140), each = 2), times = 9))`
- `dat <- data.frame(Y = y, A, B, C)`
- # Sum-to-zero (orthogonal) contrasts are default in R for unordered factors
- `fit <- aov(Y ~ A*B*C, data = dat)`
- `summary(fit)` # ANOVA table (Type I for balanced designs)

Response Surface Model

When factors are quantitative:

$$Y = \beta_0 + \sum \beta_i x_i + \sum \beta_{ii} x_i^2 + \sum \beta_{ij} x_i x_j + \varepsilon$$

This is a **second-order model**

- Linear terms $\leftrightarrow$ main effects (L)
- Quadratic terms $\leftrightarrow$ curvature (Q)
- Cross terms $\leftrightarrow$ interactions

■ **TABLE 9.3**

Regression Models for Example 9.1

Nozzle Type	$x_1 = \text{Speed (S)}, x_2 = \text{Pressure (P) in Coded Units}$
1	$\hat{y} = 22.1 + 3.5x_1 + 16.3x_2 + 51.7x_1^2 - 71.8x_2^2 + 2.9x_1x_2$ $\hat{y} = 1217.3 - 31.256S + 86.017P + 0.12917S^2 - 2.8733P^2 + 0.02875SP$
2	$\hat{y} = 25.6 - 22.8x_1 - 12.3x_2 + 14.1x_1^2 - 56.9x_2^2 - 0.7x_1x_2$ $\hat{y} = 180.1 - 9.475S + 66.75P + 0.035S^2 - 2.2767P^2 - 0.0075SP$
3	$\hat{y} = 15.1 + 20.3x_1 + 5.9x_2 + 75.8x_1^2 - 94.9x_2^2 + 10.5x_1x_2$ $\hat{y} = 1940.1 - 46.058S + 102.48P + 0.18958S^2 - 3.7967P^2 + 0.105SP$

Coded Level	Speed (psi)	Pressure (rpm)
-1	100	10
0	120	15
+1	140	20